Secondary 1
Computer Science
CA1 Term plan/ Assessment outline

This package contains

- a teaching program
- an assessment outline

Mr. Masum

Teaching program – Computer Science

Week	Unit Content	Learning Outcomes	Assessment tasks	Resources
Week 1	1. Introduction to Computer Systems -Number systems -1 homework is assigned	● Students will be able to Binary represents data ● Students will be able to learn about Binary ● Students will be learn about Denary ● Students will be learn about Hexadecimal ● Students will be able to Converting from binary to denary ● Students will be able to Converting from hexadecimal to denary and from denary to hexadecimal ● Students will be able to Use of the hexadecimal system ● Addition of binary numbers ● two's complement notation to represent positive and negative binary numbers		**Cambridge IGCSE Computer Science Coursebook pg 1-25**

Week 1-2	1. Introduction to Computer Systems -Text, Sound, and Images - Data storage and file compression **<u>Class Activities</u>** -we will convert high-quality images to low-quality images -2 handouts are provided	● How and why a computer represents text ● Representation of sound ● Representation of (bitmap) images ● sound sample rate and sample resolution how and why a computer represents an image ● the effects of the resolution and color depth on images ● Students will be able to Measurement of data storage ● Students will be able to Calculation of file size ● How Data compression works? ● Lossy and lossless file compression	Week 4	**Cambridge IGCSE Computer Science Coursebook**

Week 3	1. Introduction to Computer Systems - Types and methods of data transmission **<u>Class Activities</u>** -2 handouts are provided	● Students will be able to types and methods of data transmission how data is broken up into data packets before transmission ● packet switching (including the role of the router in the process) ● methods of data transmission (serial, parallel, simplex, half-duplex and full-duplex) ● the Universal Serial Bus (USB)	Week 6	**Cambridge IGCSE Computer Science Coursebook**

Week 4	1. Data transmission - Methods of error detection **<u>Class Activities</u>** -1 handout is provided	● Why error checking methods are needed ● Error checking methods following data transmission: parity checks, checksum and echo check ● Use of check digits to detect data entry errors ● use of automatic repeat requests (ARQs) to detect errors	Week 6	**Cambridge IGCSE Computer Science Coursebook**
Week 5	1. Data transmission - encryption **<u>Class Activities</u>** -2 handouts are provided	● The need for and the purpose of encryption ● Symmetric and asymmetric encryption ● Sse of public and private keys.		**Cambridge IGCSE Computer Science Coursebook**
Week 6	Revision			

Assessment Outline Template

Subject: Computer Science

Class: Secondary 3

Assessment type	Assessment Task	Weightings	Week
CT 1	● Number systems，Text , Sound and Images,Data storage and file compression	10%	**4**
CT 2	● Types and methods of data transmission ● methods of error detection ● encryption	10%	**6**
MCW (Formative)	● Number systems and Types and methods of data transmission	5%	**7**
Homework Submission	● convert Binary, denary and hexadecimal ● use of character sets including ASCII and Unicode ● header, payload and trailer serial, parallel, simplex, half-duplex and full-duplex ● symmetric and asymmetric encryption	5%	**Combined**

CA1	• Number systems • Text , Sound and Images • Data storage and file compression • methods of error detection • encryption	70%	**Week 9**